Coplay-Sync: A Domain-Specific Small Language Model for Physics Research Assistance with LoRA, RAG, Iterative Retrieval, and Calibrated Uncertainty

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Abstract

We present Coplay-Sync, a complete pipeline for building domain-specific Small Language Models (SLMs) for physics research assistance. Starting from a custom Rust-based Byte Pair Encoding (BPE) tokenizer with 32K vocabulary, we train a 35.2M parameter base model and apply Low-Rank Adaptation (LoRA) fine-tuning on 34,464 arXiv physics papers across 5 categories, achieving a 44% evaluation loss reduction ($0.0107 \rightarrow 0.0060$) with only 65K trainable parameters (0.19% of base). We then integrate Retrieval-Augmented Generation (RAG) with BM25 retrieval, cross-encoder reranking, and a novel 4-component calibrated uncertainty framework that reduces mean uncertainty by 33.8% and improves discrimination by 71.1%. An iterative retrieval loop with 61.76% trigger rate further refines answer quality. All components are validated through controlled ablation studies (2×2 reranker $\times$ iteration design) on a 102-question physics benchmark. The complete system operates on consumer hardware (Apple M3 Pro, 16GB RAM) and is fully open-sourced.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities across natural language processing tasks, yet general-purpose models often lack the specialized knowledge required for physics research. Physics queries demand precise mathematical reasoning, familiarity with domain-specific notation, and access to current literature. Small Language Models (SLMs) in the 3B–7B parameter range offer a compelling alternative: they are efficient to train and deploy, yet can achieve strong performance when optimized for specific domains.

This paper introduces Coplay-Sync, a complete, reproducible pipeline for building physics research assistants. Our contributions span five phases: (1) a production Rust BPE tokenizer with 32K vocabulary, (2) base model pre-training and LoRA fine-tuning on 34K+ arXiv papers, (3) comprehensive inference and evaluation infrastructure, (4) RAG integration with reranking and calibrated uncertainty, and (5) iterative retrieval with full ablation validation.

2 System Architecture

The implementation follows a streaming-to-training-to-inference architecture:

1. **Data Pipeline:** Live arXiv streaming with exponential backoff retry, multi-category collection, and automatic deduplication.

2. **Tokenizer:** Rust-based BPE with 32K vocabulary, CLI and library APIs.

3. **Training:** Tiny Transformer with MPS/CUDA/CPU auto-selection, LoRA fine-tuning, and comprehensive checkpointing.

4. **Inference:** Dual sampling profiles (production vs. canonical), streaming output, and multi-run evaluation.

5. **RAG:** BM25 retrieval, cross-encoder reranking, calibrated uncertainty, and iterative refinement.

3 Phase 1: Tokenizer and Base Model

3.1 Tokenizer

We implement byte-pair encoding (BPE) from scratch in Rust, scaling from an initial 1,000-token vocabulary to 32,000 tokens. The tokenizer handles mathematical symbols (α, β, ∇), Greek letters, subscripts/superscripts, and physics notation.

3.2 Base Model

The base model is a decoder-only Transformer with 35.2M parameters, 2,048 token context, and RMSNorm pre-normalization. Initial pre-training achieves an evaluation loss of 0.0107.

4 Phase 2: LoRA Fine-Tuning

4.1 Training Configuration

We apply Low-Rank Adaptation (LoRA) (Hu et al., 2021) with rank $r = 8$ and $\alpha = 16$ on 34,464 arXiv physics papers across 5 categories (quantum physics, optics, HEP-theory, general relativity, and general physics).

Table 1: Phase 2 Training Specifications

Component	**Value**
Corpus Size	34,464 documents (5 arXiv categories)
LoRA Rank	$r = 8$, $\alpha = 16$
Trainable Parameters	65,536 (0.19% of base)
Training Steps	10,000
Effective Batch Size	8 (4 micro-batch $\times$ 2 accumulation)
Learning Rate	0.0002 (cosine schedule)
Training Time	$\sim$10.5 hours (Apple M3 Pro, MPS)

4.2 Results

LoRA fine-tuning achieves a 44.0% evaluation loss reduction (0.0107 $\rightarrow$ 0.0060) with 99.81% parameter efficiency. All 11 checkpoints are ranked by validation loss; the best checkpoint at step 9,000 is selected for downstream tasks.

5 Phase 3: Inference and Evaluation

We implement dual sampling profiles for different use cases:

- **Production Profile:** temperature=2.0, top_k=100, max_tokens=64 (diversity-optimized)
- **Canonical Profile:** temperature=1.0, top_k=50, max_tokens=50 (reproducibility-optimized)

Table 2: Phase 1 vs Phase 2 Performance

Metric	Phase 1	Phase 2	Improvement
Eval Loss	0.0107	0.0060	↓**44.0%**
Params Trained	35.2M	65K	↓**99.81%**
Training Time	48h	10.5h	↓**78.1%**
Efficiency (loss/hour)	0.00022	0.00067	↑**3.0**×

Both profiles show consistent LoRA acceleration (>1.2× speedup) with comparable inference times (∼0.32–0.33s per prompt).

6 Phase 4: RAG Integration

6.1 Retrieval Baseline

BM25 retrieval achieves 100% accuracy on the physics test corpus. We implement retry/backoff for arXiv API resilience and automatic checkpointing for long runs.

6.2 Calibrated Uncertainty (Task 10.1)

We introduce a 4-component calibrated uncertainty framework:

Table 3: Calibrated Uncertainty Results

Metric	Before	After
Mean Uncertainty	0.662	**0.438** (↓33.8%)
Std Dev (Discrimination)	0.254	**0.073** (↓71.1%)

6.3 Cross-Encoder Fine-Tuning (Task 10.2b)

We collect 300 STEM preference pairs (24.3% positive) and fine-tune a cross-encoder reranker, achieving validation loss 0.7011 and accuracy 0.7667.

6.4 Iterative Retrieval Loop (Task 10.2c)

The iterative loop triggers on 60% of queries with an average of 1.6 iterations, refining answers through multi-step retrieval.

7 Phase 5: Full Integration and Ablation

7.1 102-Question Benchmark

The full integrated system is evaluated on 102 physics questions:

7.2 Controlled Ablation (2×2 Design)

We isolate the pure reranker effect by disabling iteration and comparing original vs. fine-tuned rerankers. The fine-tuned reranker improves rerank scores while MC exact match remains at ceiling (1.0), indicating the need for harder benchmarks.

Table 4: Phase 5 Full Benchmark Results

Metric	**Value**
MC Exact Match	1.0
Avg Calibrated Uncertainty	0.4234
Avg Iterations	1.6176
Trigger Rate	61.76%
Runtime (no iteration)	∼25 min
Runtime (with iteration)	∼41 min

8 Discussion

The current benchmark saturates at ceiling (MC exact = 1.0) on the 102-question set, suggesting the need for harder evaluation splits. Our 2-week execution plan targets: (1) 300–500 additional hard questions, (2) multi-seed significance runs ($n = 3$), (3) 1000+ strongly-labeled reranker pairs, and (4) human evaluation for faithfulness and helpfulness.

9 Conclusion

Coplay-Sync demonstrates that domain-specific SLMs can deliver research-grade performance at a fraction of the computational cost of general-purpose LLMs. The complete pipeline—from tokenizer to calibrated RAG—is reproducible on consumer hardware and fully open-sourced.

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References

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